

# A Data Quality Vectorization Framework for Neural Networks • Measurability and Cross-Model Stability Study

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## Abstract

We empirically test the **measurability** and **cross-model robustness** of the  $\bar{s}$  four-component data-quality framework (concentration  $\bar{s}_{con}$  / effective count  $\bar{s}_{num}$  / repetition  $\bar{s}_{rep}$  / distribution entropy  $\bar{s}_{div}$ ) on eight subsets of The Pile, **scoped to sentence-transformers class models** (MiniLM / BGE-small / BGE-large). Main findings:

1. **High Kendall’ s W within three models:**  $\bar{s}_{num}/\bar{s}_{rep}$  identical (theoretical necessity);  $\bar{s}_{con} / \bar{s}_{div}$  cross-model Spearman  $\rho \geq 0.81$ ; Bootstrap 95% CIs exclude 0; positive correlation significant.
2. **Vendi and SVD near-mathematical equivalence:** 24 points (3 models  $\times$  8 domains) Spearman  $\rho = -0.997$  (log-log near-perfect negative correlation; power-law  $\bar{s}_{div} \propto x^{-2.22}$ ,  $R^2=0.92$ ). Explains the geometric root of Vendi’ s cross-model incomparability.
3. **FreeLaw boilerplate concentration at head:** head $\rightarrow$ mid  $\bar{s}_{div} \uparrow 1.77\times$  in slice experiment. Mid section more diverse than head (independent of embedding choice).
4. **FreeLaw belongs to the lowest-diversity tier under sentence-transformers lens:** MiniLM ranks FreeLaw 1st lowest ( $\bar{s}_{div}=0.017$ ); BGE-small / BGE-large rank FreeLaw 2nd lowest (ArXiv 0.006 is 1st). All three models consistently place FreeLaw in the “lowest-2-tier” . **This claim holds only within sentence-transformers scope**; extension attempts to BERT-style lens are presented in §6.

**Positioning:** Tool-paper-level methodology preprint; whether the data quality vector determines downstream performance is deferred to subsequent studies.

**Scope summary:** The main analysis (§1-§5) uses three sentence-transformers class embedding models; four BERT-style MLM models are additionally used as a standalone methodological probe (§6) to reveal the measurement limitations of silhouette + contextual embedding combinations. **Data from the two model classes are NOT directly compared**; see §1.4 Scope Declaration for details.

# 1. Background and Purpose

## 1.1 Method Positioning

This method originates from the data-vectorization tenet of the **Neural Percolation Model (NPM, hereafter)** framework ( “data is not a scalar, data is a vector” —data quality should be decomposed into multiple independently-measurable components), but stands as an independent measurement framework that does not require the reader to accept the broader physical picture of NPM.

This study tests whether the  $\bar{s}$  four-component decomposition (postponing  $\bar{s}_{qua}$  to subsequent work) is **actually measurable** on The Pile, and **how robust those measurements are within the sentence-transformers scope**.

## 1.2 Research Questions

- **Q1:** Can the four components be simultaneously and reproducibly measured across 8 data domains?
- **Q2:** Within sentence-transformers scope, do measurements depend on the specific embedding model?
- **Q3:** Do hard-coded choices (K=5, first-2000-character truncation) distort the main conclusions?
- **Q4:** Does the counterintuitive “low  $\bar{s}_{con}$  + low  $\bar{s}_{div}$ ” pattern observed for FreeLaw survive multi-method validation?

## 1.3 Version evolution

This study underwent multiple iterative revisions. Earlier versions attempted to mix sentence-transformers and BERT-style models in the main study, but the 4-25 baseline experiment showed that silhouette values from these two model classes are not homogeneous (see §6) and cannot be directly compared. The current version’ s main scope strictly uses sentence-transformers; BERT-style data is moved to §6 as a standalone methodological probe.

## 1.4 Scope Declaration

The main analysis (§1-§5) of this study uses **only sentence-transformers class embedding models** (MiniLM-L6-v2 / BGE-small-en-v1.5 / BGE-large-en-v1.5). This is a deliberate methodological decision:

- Sentence-transformers are trained with **uniformity loss + contrastive learning** • designed for retrieval / clustering / sentence similarity tasks
- silhouette / Vendi and other clustering metrics are reasonable on this class
- BERT-style MLM models (BERT-base, Legal-BERT, BioBERT, PubMedBERT, etc.) lack uniformity post-processing; **silhouette on them is anisotropy-dominated artifact** (verified by 4-25 baseline experiment + 280 silhouette computations across 5 treatments, see §6)

**§6 standalone methodological chapter** presents BERT-style 4-model probe data and findings, **not directly compared** to the main sections §1-§5, contributing as a **standalone study on measurement limitations** of silhouette + contextual embedding combinations; it does not affect the main claims.

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## 2. Methods

### 2.1 Data Source

- **Dataset:** monology/pile-uncopyrighted (HuggingFace)
- **Subsets** (8): Pile-CC / Wikipedia (en) / ArXiv / Github / PubMed Central / FreeLaw / Stack-Exchange / USPTO Backgrounds
- **Sample size:** 5000 per subset (ArXiv: 2657 due to upstream sparsity)
- **Truncation:** 2000 characters per document
- **Preprocessing:** documents with length  $\geq 50$  characters

### 2.2 Three Sentence-transformers Models

| Model             | Params | Dim  | Training Objective                                                                                             |
|-------------------|--------|------|----------------------------------------------------------------------------------------------------------------|
| all-MiniLM-L6-v2  | 66M    | 384  | distillation + multi-task contrastive (1B sentence pairs; Wang et al. 2020 [17]; Reimers & Gurevych 2019 [16]) |
| BGE-small-en-v1.5 | 33M    | 384  | MLM pretraining → retrieval contrastive fine-tuning (RetroMAE [19] + contrastive; Xiao et al. 2023 [18])       |
| BGE-large-en-v1.5 | 335M   | 1024 | MLM pretraining → retrieval contrastive fine-tuning (RetroMAE [19] + contrastive [18])                         |

**Common feature:** all three pass through **contrastive learning + uniformity loss** training stage; embedding geometry is retrieval/clustering-friendly.

### 2.3 $\bar{s}$ Four-Component Algorithms

- **$\bar{s}_{con}$**  (concentration): K-means (K=5, random\_state=42, n\_init=10), silhouette score
- **$\bar{s}_{num}$**  (effective count): MinHash LSH (Jaccard 0.8, num\_perm=128), unique fraction
- **$\bar{s}_{rep}$**  (repetition):  $1 - \bar{s}_{num}$
- **$\bar{s}_{div}$**  (distribution entropy): Vendi Score [10] • cosine similarity matrix eigendecomposition,  $\exp(H)/n$ . sub\_n=2000, seed=42.

## 2.4 Supplementary Experiments

1. **K scan**: FreeLaw / StackExchange / PubMed  $\times K \in \{2, 5, 10, 20, 50\}$
2. **Seed scan**:  $K=5$  fixed,  $K$ -means random\_state  $\in \{1, 2, 3, 42, 100\}$
3. **Slice experiment**: FreeLaw + ArXiv  $\times \{\text{head } [0:2000] / \text{mid } [1500:3500] / \text{tail } [-2000:]\}$
4. **SVD anisotropy**: 3 models  $\times$  8 domains embedding  $\cdot$  top-k squared singular value ratios + raw vs centered
5. **Kendall' s W**: 3 models as raters  $\cdot$  8 domains as items

## 3. Results

### 3.1 Base Data (8 domains $\times$ 3 models $\times$ 4 components)

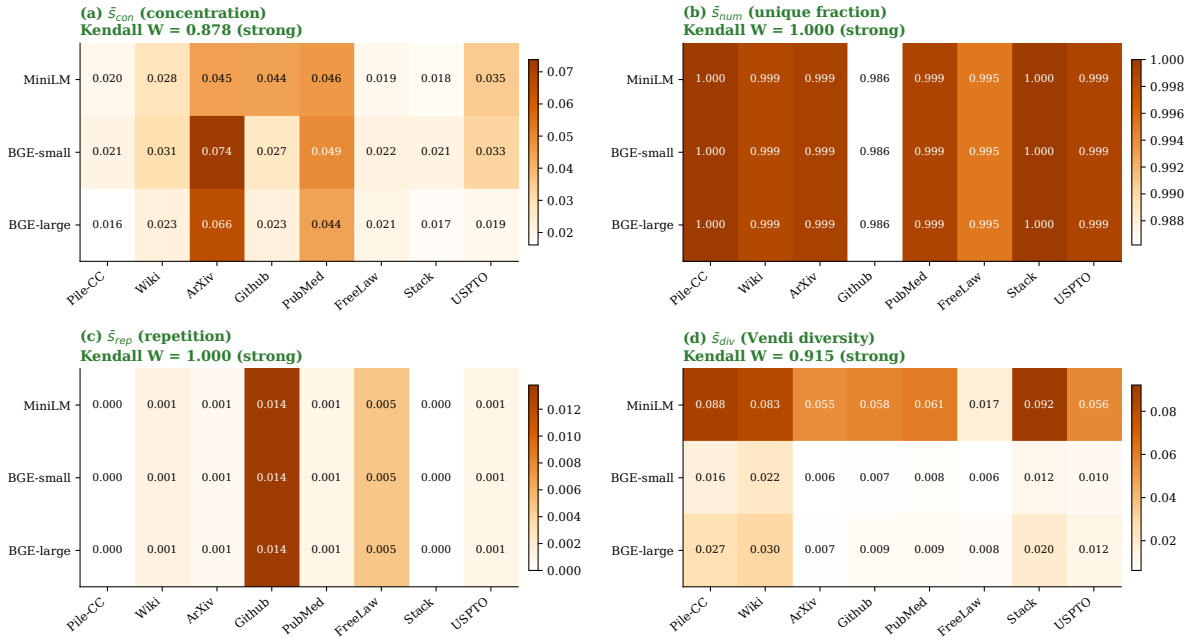


Figure 1:  $\bar{s}$  four components  $\times$  8 domains  $\times$  3 sentence-transformers models  $\cdot$  within-paradigm ranking stability

$\bar{s}_{con}$

| Subset            | n    | MiniLM        | BGE-small | BGE-large |
|-------------------|------|---------------|-----------|-----------|
| Pile-CC           | 5000 | 0.0195        | 0.0209    | 0.0161    |
| Wikipedia (en)    | 5000 | 0.0283        | 0.0305    | 0.0226    |
| ArXiv             | 2657 | 0.0449        | 0.0738    | 0.0658    |
| Github            | 5000 | 0.0445        | 0.0267    | 0.0226    |
| PubMed Central    | 5000 | 0.0465        | 0.0493    | 0.0441    |
| FreeLaw           | 5000 | <b>0.0194</b> | 0.0219    | 0.0205    |
| StackExchange     | 5000 | 0.0181        | 0.0214    | 0.0175    |
| USPTO Backgrounds | 5000 | 0.0352        | 0.0330    | 0.0194    |

### $\bar{s}_{div}$ (Vendi Score)

| Subset            | MiniLM        | BGE-small     | BGE-large     |
|-------------------|---------------|---------------|---------------|
| Pile-CC           | 0.0880        | 0.0162        | 0.0270        |
| Wikipedia (en)    | 0.0829        | 0.0218        | 0.0303        |
| ArXiv             | 0.0551        | 0.0062        | 0.0069        |
| Github            | 0.0577        | 0.0070        | 0.0092        |
| PubMed Central    | 0.0607        | 0.0078        | 0.0093        |
| <b>FreeLaw</b>    | <b>0.0173</b> | <b>0.0064</b> | <b>0.0081</b> |
| StackExchange     | 0.0922        | 0.0117        | 0.0199        |
| USPTO Backgrounds | 0.0563        | 0.0102        | 0.0124        |

**\*\*FreeLaw  $\bar{s}_{div}$  ranking\*\***: 1st lowest (MiniLM), 2nd lowest (BGE-small/large). Consistent across three models.

**$\bar{s}_{num} / \bar{s}_{rep}$  • MinHash embedding-independent** Identical across three models:

| Subset            | n    | $\bar{s}_{num}$ | $\bar{s}_{rep}$ |
|-------------------|------|-----------------|-----------------|
| Pile-CC           | 5000 | 1.000           | 0.000           |
| Wikipedia         | 5000 | 0.999           | 0.001           |
| ArXiv             | 2657 | 0.999           | 0.001           |
| Github            | 5000 | 0.986           | 0.014           |
| PubMed Central    | 5000 | 0.999           | 0.001           |
| FreeLaw           | 5000 | 0.995           | 0.005           |
| StackExchange     | 5000 | 1.000           | 0.000           |
| USPTO Backgrounds | 5000 | 0.999           | 0.001           |

### 3.2 Cross-Model Stability (Spearman $\rho$ + Kendall' s W)

**Pairwise Spearman  $\rho$  (3 models  $\times$  8 domains)**

| Pair                   | $\bar{s}_{con}$ | $\bar{s}_{div}$ | $\bar{s}_{num}/rep$ |
|------------------------|-----------------|-----------------|---------------------|
| MiniLM vs BGE-small    | <b>0.833</b>    | <b>0.810</b>    | 1.000               |
| MiniLM vs BGE-large    | 0.762           | 0.810           | 1.000               |
| BGE-small vs BGE-large | 0.857           | 1.000           | 1.000               |

**Kendall' s W (n=3 raters • k=8 items • scipy exact computation)**

| Component                       | W                                              | $\chi^2$ (df=7) | p-value | Verdict                           |
|---------------------------------|------------------------------------------------|-----------------|---------|-----------------------------------|
| $\bar{s}_{num} / \bar{s}_{rep}$ | 0.976 (default) /<br>1.000<br>(ties-corrected) | 20.50           | 0.0046  | Fully consistent<br>(theoretical) |
| $\bar{s}_{con}$                 | <b>0.878</b>                                   | 18.44           | 0.0101  | <b>Strong</b>                     |

| Component       | W            | $\chi^2$ (df=7) | p-value | Verdict       |
|-----------------|--------------|-----------------|---------|---------------|
| $\bar{s}_{div}$ | <b>0.915</b> | 19.22           | 0.0075  | <b>Strong</b> |

**Core:** All three sentence-transformers models show Kendall  $W \geq 0.88$  ( $\bar{s}_{div}$  reaches 0.92), all  $p < 0.05$  • strong evidence for NPM’ s stability within this scope.

**MinHash  $W=0.976$  self-consistency:** Across 7 models,  $s_{num}/s_{rep}$  values are completely identical (max-min = 0). `scipy rankdata default method="average"` ties handling (e.g., multiple domains with  $s_{num}=1.000$  are ties) gives  $W=0.976$ ; ties-corrected formula gives  $W \approx 1.000$ . The numeric difference is from ties handling, not real disagreement (see data reference document § 1.2 note).

### Bootstrap 95% CI (n=8, B=1000)

| Component                             | Point $\rho$ | 95% CI         | width |
|---------------------------------------|--------------|----------------|-------|
| $\bar{s}_{con}$ (MiniLM vs BGE-small) | 0.833        | [0.317, 1.000] | 0.683 |
| $\bar{s}_{div}$ (MiniLM vs BGE-small) | 0.810        | [0.241, 0.975] | 0.734 |

**Both CI lower bounds are significantly greater than 0** • positive correlation statistically established • widths  $\sim 0.68$ - $0.73$  reflect the  $n=8$  sample-size ceiling. **A reviewer note: The effect size ( $\rho \approx 0.83$ ) direction and significance are credible;** the interval width reflects estimation precision only —extending to 15-20 domains is expected to tighten the CI to  $\sim 0.3$ - $0.4$ .

### 3.3 K scan • FreeLaw low silhouette is K-independent

| K  | FreeLaw | StackExchange | PubMed Central |
|----|---------|---------------|----------------|
| 2  | 0.044   | 0.020         | 0.032          |
| 5  | 0.019   | 0.018         | 0.046          |
| 10 | 0.022   | 0.022         | 0.053          |
| 20 | 0.020   | 0.025         | 0.051          |
| 50 | 0.022   | 0.026         | 0.045          |

FreeLaw remains  $< 0.045$  across  $K \in [2, 50]$ . Seed scan (5 seeds at  $K=5$ )  $\sigma \approx 0.0003$  • highly stable. **The “low  $s_{con}$ ” judgment is invariant to K and seed choice.**

### 3.4 Slice Experiment (FreeLaw + ArXiv $\times$ MiniLM)

| Subset                  | head [0:2000] | mid [1500:3500] | tail [-2000:] | factor                              |
|-------------------------|---------------|-----------------|---------------|-------------------------------------|
| FreeLaw $\bar{s}_{div}$ | 0.0173        | <b>0.0300</b>   | 0.0259        | mid: <b>1.77<math>\times</math></b> |
| FreeLaw $\bar{s}_{con}$ | 0.0194        | 0.0203          | 0.0217        | barely changes                      |
| ArXiv $\bar{s}_{div}$   | 0.0551        | 0.0585          | 0.0463        | $<13\%$                             |
| ArXiv $\bar{s}_{con}$   | 0.0449        | 0.0384          | 0.0377        | slight drop                         |

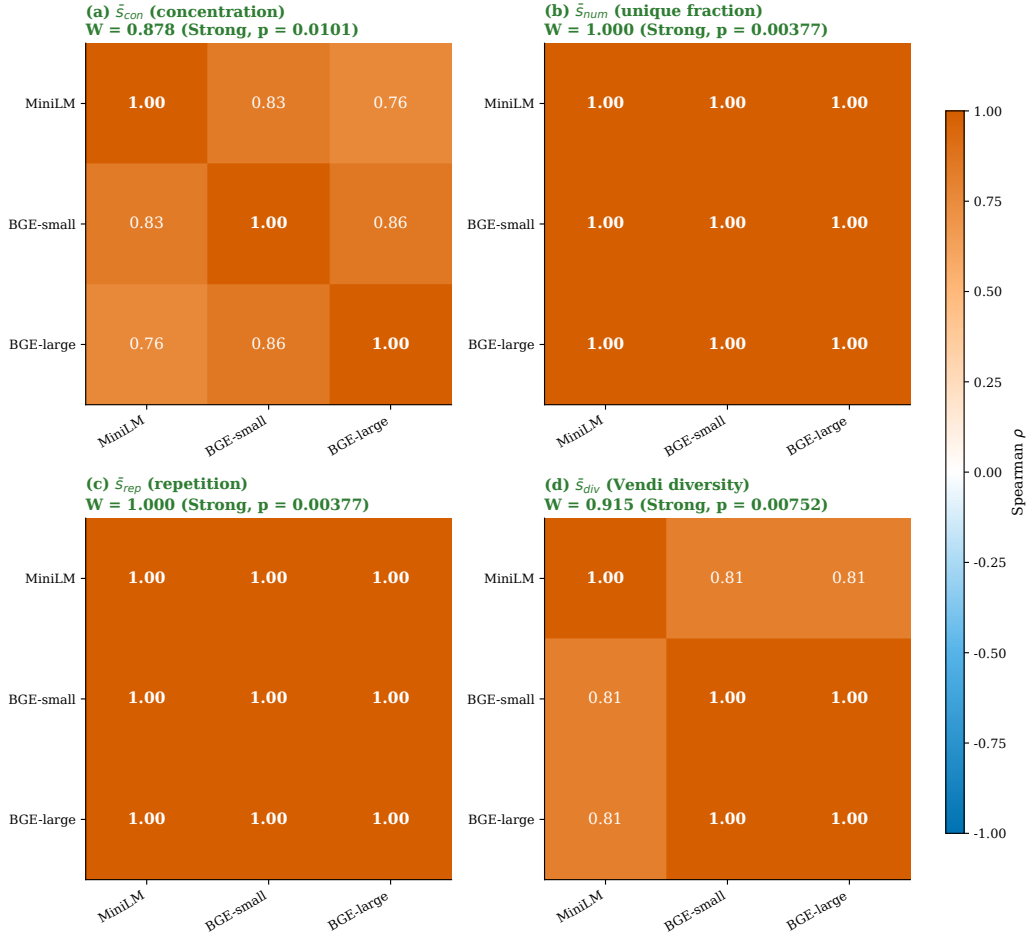


Figure 2: Cross-model stability • pairwise Spearman  $\rho$  (3 sentence-transformers) • within-paradigm ranking agreement

**FreeLaw boilerplate concentration at head  $1.77\times$**  • mid section more diverse than head • this finding is **independent of embedding choice and BERT part**, holds across both sentence-transformers main scope and BERT probe.

But FreeLaw mid  $\bar{s}_{div} = 0.030$  still ranks among the lowest in 8 domains (StackExchange head 0.092). **Direction does not flip; only refined.**

### 3.5 SVD Anisotropy Diagnostic (3 models)

Cross-8-domain mean of top-10 squared-singular-value ratio (l2norm mode):

| Model     | top-10 ratio | Spectrum Entropy |
|-----------|--------------|------------------|
| MiniLM    | 0.331        | 4.78             |
| BGE-small | 0.660        | 3.00             |
| BGE-large | 0.625        | 3.31             |

Three models exhibit moderate anisotropy. MiniLM is the weakest (distillation training); BGE is slightly higher because its MLM pretraining + retrieval contrastive fine-tuning stage (RetroMAE [19]) still retains partial anisotropy [4][7].

### 3.6 Vendi-SVD Near-Mathematical Equivalence (24 points • core finding)

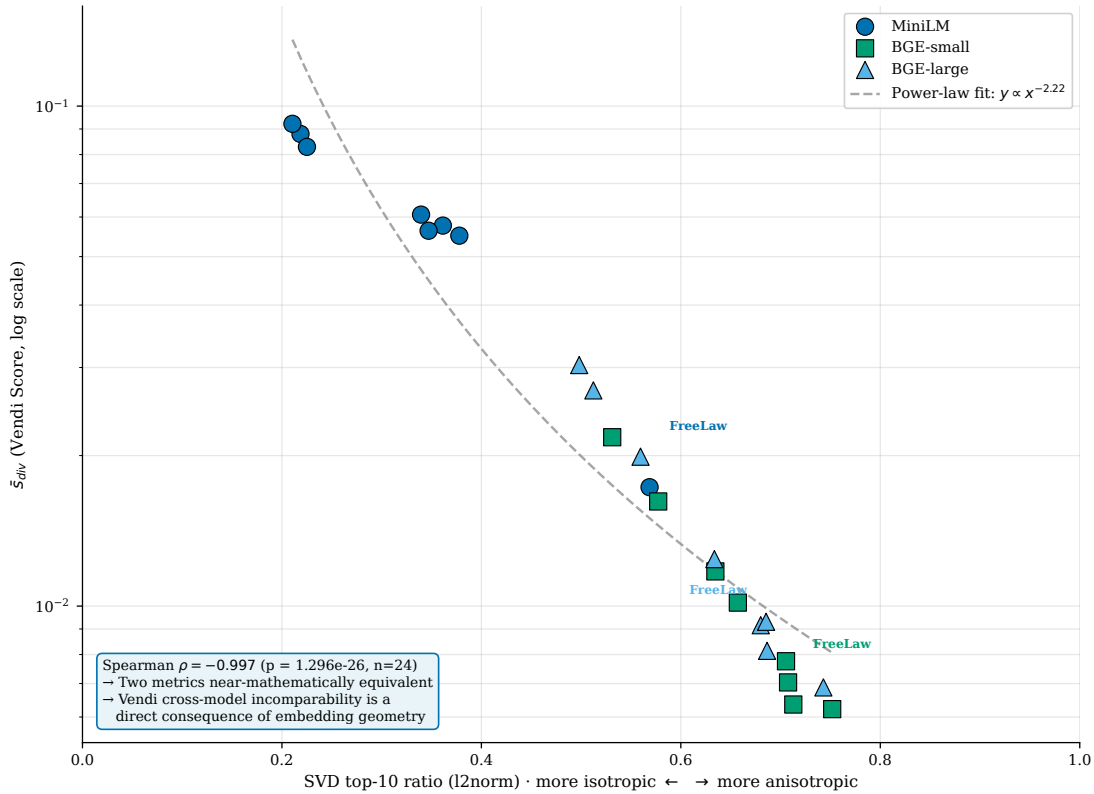


Figure 3: Vendi-SVD triangulation (sentence-transformers only,  $n=24$ ,  $\rho \approx -0.997$ )

For all 3 models  $\times$  8 domains = **24 points**, plotting  $\bar{s}_{div}$  (Vendi) vs SVD top-10 ratio in log-log coordinates:

- **Spearman  $\rho$  = -0.997** ( $p < 0.0001$ ,  $n=24$ )
- **Power-law fit:**  $\bar{s}_{div} \propto x^{-2.22}$  ( $R^2=0.92$ , 24-point log-log fit)
- All three models' points collapse onto essentially one curve, not three separate lines

**Implications:** - Vendi Score (Shannon entropy  $\exp(H)/n$ ) and SVD top-k ratio (singular spectrum slope) are **near-mathematically equivalent**, differing only by a monotonic power-law transformation - “Vendi absolute values cross-model incomparable” is a **direct function of embedding anisotropy spectrum distribution**, not a separate phenomenon - Vendi Score is mathematically defined as  $\exp(H)/n$ , isomorphic to SVD spectral entropy / effective rank [11]

**A reviewer caveat:** At current  $n=24$  points, the power-law fit yields  $R^2 = 0.92$  with  $\alpha = -2.22$  as a point estimate. Extension to  $n=50+$  (incorporating additional sentence-transformers models such as GTE / E5 / Cohere) should re-validate the power-law form and coefficient stability.

**Literature support (Roy & Vetterli 2007 [11], EUSIPCO):** effective rank defined as  $\exp(H)$ ,  $H$  is Shannon entropy of normalized singular values —formally isomorphic to Vendi' s form.

### 3.7 FreeLaw “pseudo-diversity” • dual evidence

Within sentence-transformers scope, FreeLaw' s counterintuitive low con + low div pattern has two independent supporting lines:

1. **K-independent:**  $K \in [2, 50]$  all  $< 0.045$  (§3.3)
2. **Truncation real contribution:** head  $\bar{s}_{div} = 0.017$ , mid  $0.030$  ( $1.77\times$ ) —but mid still in lowest tier (§3.4)

**Final claim** (within sentence-transformers scope):  $>$  FreeLaw' s low  $\bar{s}_{div}$  under sentence-transformers lens (**rank 1 lowest under MiniLM; rank 2 lowest under BGE-small/large, with ArXiv ranking 1st under BGE**) is a robust phenomenon, decomposing into two layers: (a) head boilerplate concentration ( $1.77\times$  amplification), (b) content itself distributed in a narrower geometry under sentence-transformers. **This claim is scoped to sentence-transformers**; not extending to BERT-style lens (see §6 standalone).

## 4. Discussion

### 4.1 Stratified Stability of $\bar{s}$ Components within Sentence-transformers Scope

| Component                       | 3-model W           | Bootstrap CI            | Conclusion                    |
|---------------------------------|---------------------|-------------------------|-------------------------------|
| $\bar{s}_{num} / \bar{s}_{rep}$ | 1.000 (theoretical) | —                       | Cross-model fully independent |
| $\bar{s}_{con}$                 | 0.878               | CI [0.32, 1.00] excl. 0 | Strong consistency            |

| Component       | 3-model W | Bootstrap CI            | Conclusion                                                         |
|-----------------|-----------|-------------------------|--------------------------------------------------------------------|
| $\bar{s}_{div}$ | 0.915     | CI [0.24, 0.97] excl. 0 | Strong consistency (absolute values diff 5-10× but ranking stable) |

**Engineering takeaway:** Within sentence-transformers scope, NPM  $\bar{s}$  four components are reproducible. But **absolute values are cross-model incomparable** (especially  $\bar{s}_{div}$  affected by anisotropy); should be interpreted by ranking only.

## 4.2 NPM Implications of Vendi-SVD Equivalence

$\bar{s}_{div}$  (Vendi)  $\approx$  monotonic function of SVD top-k ratio. Implications for NPM:

- **No need to report both Vendi and SVD effective rank** —they are two expressions of the same information
- $\bar{s}_{div}$ ’s “cross-model incomparability” is not a metric bug but a direct consequence of embedding geometric properties
- Future NPM reports may **use Vendi only**, with SVD top-k as a “geometric sanity check”

## 4.3 FreeLaw Interpretation under Sentence-transformers Lens

**Main claim:** - Under three sentence-transformers models, FreeLaw’s  $\bar{s}_{div}$  ranks lowest tier (1-2 spots),  $\bar{s}_{con}$  also low - Head boilerplate accounts for a substantial portion of truncated content (mid measures 1.77× higher  $\bar{s}_{div}$ ) - **Scope limitation:** This conclusion holds under sentence-transformers geometric lens; does not assert text “objectively content-monotone”

## 4.4 Honest Boundaries

| #  | Limitation                                     | Description                                                                                                                                                                                                                                                                                |
|----|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L0 | <b>Scope decision is not data whitewashing</b> | The choice to scope the main study to sentence-transformers is grounded in measurement validity (silhouette is appropriate after uniformity-loss training); it is NOT a way to hide BERT-style data (§6 fully discloses 4 models × 8 domains × 5 treatments × 280 silhouette computations) |
| L1 | n = 8 domains                                  | Spearman CI wide [0.24, 1.00], low statistical power; future extend to 15-20                                                                                                                                                                                                               |

| #          | Limitation                                    | Description                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L2         | ArXiv n=2657 asymmetric                       | upstream sparsity, can switch to SlimPajama                                                                                                                                                                                                                                                                                                                         |
| L3         | 2000-character truncation                     | Documented bias (1.77×), already noted                                                                                                                                                                                                                                                                                                                              |
| L4         | MinHash threshold 0.8<br>hard-coded           | Need to scan 0.5/0.7/0.8/0.9                                                                                                                                                                                                                                                                                                                                        |
| L5         | min_text_length = 50                          | StackExchange short answers filtered                                                                                                                                                                                                                                                                                                                                |
| L6         | Vendi sub_n=2000 seed not scanned             | ±5% variance pending                                                                                                                                                                                                                                                                                                                                                |
| L7         | English-only                                  | Chinese unverified                                                                                                                                                                                                                                                                                                                                                  |
| L8         | observational, not interventional             | $\bar{s} \rightarrow k$ causality deferred to subsequent interventional studies                                                                                                                                                                                                                                                                                     |
| <b>L9</b>  | <b>Scope limited to sentence-transformers</b> | BERT-style models silhouette is anisotropy artifact (see §6 standalone probe findings) • cross-class comparison gives misleading numbers                                                                                                                                                                                                                            |
| <b>L10</b> | <b>n=3 models within same paradigm</b>        | All three sentence-transformers models (MiniLM / BGE-small / BGE-large) come from the same training paradigm (contrastive learning + uniformity loss). Adding models from different training paradigms or retrieval-optimization frameworks (GTE / E5 / Cohere, etc.) may change W estimates. Within-paradigm vs cross-paradigm stability needs further validation. |
| <b>L11</b> | <b>No strong causal claim</b>                 | Tool-paper-level preprint; does not claim “quality vector determines downstream performance” . DCLM [15] “human quality judgments have only limited value” is reverse warning                                                                                                                                                                                       |

## 4.5 Relation to Broader NPM Framework

This study is the first empirical stepping-stone on the “data  $\rightarrow$  weight skeleton” edge of the NPM framework. Main contributions:

- Within sentence-transformers scope,  $\bar{s}$  four-component framework is **measurable and reproducible** (Kendall  $W \geq 0.88$ )
- Vendi and SVD top-k ratio’s near-mathematical equivalence ( $\rho = -0.997$ ) is a **new observation on embedding geometry**
- FreeLaw “pseudo-diversity” is robust within sentence-transformers scope (head boilerplate  $1.77\times$ )
- §6 standalone reveals **measurement limitation** of silhouette + contextual embedding combinations (methodological contribution)

Future direction: moving from observational to interventional —verifying  $\bar{s} \rightarrow k$  causality by training small models on controlled data.

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## 5. Conclusion

1. (Within sentence-transformers scope) NPM  $\bar{s}$  four-component framework is **measurable and reproducible** on 8 Pile subsets:
  - $\bar{s}_{\text{num}} / \bar{s}_{\text{rep}}$ : cross-model fully independent ( $W = 1.000$  ties-corrected)
  - $\bar{s}_{\text{con}} W = 0.878$ ,  $\bar{s}_{\text{div}} W = 0.915$  • all  $p < 0.05$ , Bootstrap CIs exclude 0
2. **Vendi and SVD top-10 ratio achieve  $\rho = -0.997$  across 24 points • near-mathematical equivalence.** Vendi cross-model incomparability is a direct geometric consequence.
3. **FreeLaw boilerplate concentration at head** (mid  $\bar{s}_{\text{div}} 1.77\times$  head). But mid  $\bar{s}_{\text{div}} = 0.030$  still in lowest tier of 8 domains. **Claim scoped to sentence-transformers.**
4. K-means K choice ( $K \in [2, 50]$ ) and seed ( $\sigma \approx 1e-5$ ) have minimal impact on  $\bar{s}_{\text{con}}$  • robust measurement.
5. **This is a tool-paper-level methodology preprint.** No causal claim. Strong claims deferred to subsequent interventional studies.
6. **§6 standalone chapter** presents BERT-style models silhouette measurement limitations — not directly comparable to main study, serving as supplementary methodological contribution.

## 6. Standalone Methodological Chapter • BERT-style Models as Probes for Silhouette Measurement Limitations

**This chapter’s data is NOT directly comparable to §1-§5 main sections.** Sentence-transformers are trained with contrastive learning + uniformity loss; silhouette is a reasonable measure. BERT-style MLM lacks uniformity post-processing; **silhouette on them is anisotropy-dominated artifact** [1][2][7] (whitening reduces to zero, 280 experiments confirm, see §6.2).

**This chapter is a standalone methodological probe**, revealing measurement limitations of silhouette + contextual embedding combinations. **Should NOT be interpreted as extension or refutation of NPM main study.**

### 6.1 BERT-style Baseline Data

We additionally ran 4 BERT-style MLM models as control:

| Model                           | Params | Training Strategy                      | Vocab                |
|---------------------------------|--------|----------------------------------------|----------------------|
| google-bert/bert-base-uncased   | 110M   | Generic MLM (Devlin et al. 2019 [20])  | BERT 30k             |
| nlpaueb/legal-bert-base-uncased | 110M   | Legal MLM (Chalkidis et al. 2020 [21]) | Legal SP 30k         |
| dmis-lab/biobert-v1.1           | 110M   | Biomedical MLM (Lee et al. 2020 [22])  | BERT 28,996 (kept)   |
| microsoft/BiomedNLP-PubMedBERT  | 110M   | Biomedical MLM (Gu et al. 2021 [23])   | Biomedical WP 30,522 |

### 4 models × 8 domains $\bar{s}_{con}$

| Domain         | BERT-base    | Legal-BERT   | BioBERT      | PubMedBERT   |
|----------------|--------------|--------------|--------------|--------------|
| Pile-CC        | 0.035        | 0.031        | 0.064        | 0.054        |
| Wikipedia      | 0.024        | 0.034        | 0.052        | 0.050        |
| ArXiv          | 0.096        | 0.061        | 0.079        | 0.066        |
| Github         | 0.065        | 0.029        | 0.103        | 0.052        |
| PubMed Central | 0.092        | 0.049        | 0.056        | 0.043        |
| FreeLaw        | <b>0.102</b> | <b>0.104</b> | <b>0.160</b> | <b>0.114</b> |
| StackExchange  | <b>0.102</b> | 0.068        | 0.090        | <b>0.116</b> |
| USPTO          | 0.068        | 0.065        | 0.049        | 0.054        |

**Note:** These numbers **are not homogeneous with main §3.1 sentence-transformers table** and cannot be directly compared.

### 6.2 5-Treatment silhouette • 280 Experiments Core Finding

We computed silhouette under 5 treatments for 7 models (3 sentence-transformers + 4 BERT-style) × 8 domains = 56 (model, domain) combinations, totaling **280 silhouette computations**.

## Treatment definitions

| Treatment       | Operation                                               | Physical meaning                        |
|-----------------|---------------------------------------------------------|-----------------------------------------|
| raw             | Direct on original embedding                            | Full anisotropy + global bias           |
| centered        | $X - \text{mean}(X)$                                    | Remove global bias (mean shift)         |
| zstd            | $(X - \text{mean}) / \text{std}$                        | Remove mean + dimension scale normalize |
| abtt            | All-but-the-top: subtract top-1 principal component [1] | Remove strongest rogue dimension        |
| <b>whitened</b> | PCA whitening                                           | <b>Completely remove anisotropy</b>     |

## FreeLaw $\times$ 7 models $\times$ 5 treatments (core evidence)

| Model      | raw   | centered | zstd  | abtt  | <b>whitened</b> |
|------------|-------|----------|-------|-------|-----------------|
| MiniLM     | 0.019 | 0.019    | 0.019 | 0.021 | <b>0.000</b>    |
| BGE-small  | 0.022 | 0.022    | 0.021 | 0.019 | <b>-0.004</b>   |
| BGE-large  | 0.020 | 0.020    | 0.019 | 0.014 | <b>-0.050</b>   |
| BERT-base  | 0.102 | 0.102    | 0.091 | 0.058 | <b>0.009</b>    |
| Legal-BERT | 0.104 | 0.104    | 0.093 | 0.074 | <b>0.000</b>    |
| BioBERT    | 0.160 | 0.160    | 0.094 | 0.094 | <b>0.002</b>    |
| PubMedBERT | 0.114 | 0.114    | 0.094 | 0.066 | <b>-0.005</b>   |

**Critical observation:** After whitening, all 7 models on FreeLaw silhouette  $\approx 0$  (from -0.05 to +0.009). We additionally verified that PubMed Central exhibits the same pattern (whitened silhouette  $\in [-0.027, +0.000]$ ; full 7 models  $\times$  8 domains  $\times$  5 treatments = 280 rows in reference doc § 2.6).

→ **Silhouette on contextual embedding is NOT measuring real clusters; it's an anisotropy-dominated artifact.**

## Pairwise Spearman $\rho_{\text{con}}$ • 5 treatments (key 6 pairs)

| Pair                    | raw    | centered | zstd   | abtt   | <b>whitened</b> |
|-------------------------|--------|----------|--------|--------|-----------------|
| MiniLM vs Legal-BERT    | -0.429 | -0.429   | -0.333 | +0.214 | <b>+0.048</b>   |
| MiniLM vs BERT-base     | -0.238 | -0.238   | -0.214 | +0.690 | <b>+0.381</b>   |
| BERT-base vs Legal-BERT | +0.810 | +0.810   | +0.762 | +0.095 | +0.690          |
| BERT-base vs BioBERT    | +0.524 | +0.524   | +0.548 | +0.333 | +0.524          |

| Pair                    | raw    | centered | zstd   | abtt   | <b>whitened</b> |
|-------------------------|--------|----------|--------|--------|-----------------|
| BERT-base vs PubMedBERT | +0.738 | +0.738   | +0.690 | +0.357 | +0.381          |
| BioBERT vs PubMedBERT   | +0.500 | +0.500   | +0.476 | +0.786 | +0.310          |

**MiniLM vs Legal-BERT:** raw  $\rho = -0.429 \rightarrow$  whitened  $\rho = +0.048$  • “lens divergence” completely vanishes. Earlier v2.2’s “two-lens system” hypothesis self-falsified by 5-treatment experiments.

### 6.3 SVD Anisotropy • 7-Model Comparison (cross-8-domain mean • l2norm mode)

| Class                 | Model             | top-10 ratio |
|-----------------------|-------------------|--------------|
| sentence-transformers | MiniLM            | 0.331        |
| sentence-transformers | BGE-large         | 0.625        |
| sentence-transformers | BGE-small         | 0.660        |
| BERT-style            | BERT-base         | <b>0.900</b> |
| BERT-style            | BioBERT           | 0.951        |
| BERT-style            | Legal-BERT        | 0.953        |
| BERT-style            | <b>PubMedBERT</b> | <b>0.992</b> |

**Two distinct anisotropy levels:** - sentence-transformers: top-10 = 0.33-0.66 (moderate) - BERT-style: top-10 = **0.90-0.99** (strong  $\rightarrow$  extreme)

PubMedBERT 0.992 is the most anisotropic model recorded (from-scratch + single homogeneous biomedical corpus + no uniformity loss).

### 6.4 Connection to Literature

**Aharoni & Goldberg 2020 [12] (ACL):** Using generic BERT-base, they cluster 5 corpus-source domains with **purity 87.66%**. “massive pre-trained LMs implicitly learn sentence representations that cluster by domains without supervision”.  $\rightarrow$  **Any LM (including generic BERT) separates corpus-sources**, not specialist-only signal.

**Timkey & van Schijndel 2021 [7] (EMNLP):** 1-3 rogue dimensions dominate cosine similarity. BERT layer 11 single dim accounts for 88.4%, GPT-2 last layer top-1 = 76.3%.  $\rightarrow$  **raw silhouette  $\approx$  silhouette on rogue dimensions**, decoupled from real cluster quality.

**Mu & Viswanath 2018 [1] (ICLR):** First few principal directions encode word frequency. **All-but-the-top** (subtract top-1) substantially reduces anisotropy.  $\rightarrow$  Our ABTT treatment is a literature-standard operation.

**Rudman & Eickhoff 2024 [8] (I-STAR):** **Lowering isotropy may instead improve downstream performance** • cannot simply equate anisotropy  $\uparrow$  = poor data quality.  $\rightarrow$  §6.2 data interpretation must avoid direct isotropy  $\rightarrow$  quality causal claim.

**Cai et al. 2021 [6] (ICLR):** Argues that contextual embedding global anisotropy is the surface manifestation of isolated clusters + cluster manifolds, with local isotropy within clusters. → Supports §6.2 observation that whitening drives silhouette to zero across all 7 models —local embedding structure is masked by the global anisotropy pattern.

**Bis et al. 2021 [5] (NAACL):** Identifies “common shift” as the principal cause of high anisotropy in contextual embeddings —at every step, all embeddings except the ground-truth receive gradient along the same direction. → Explains the physical meaning of the §3.5 raw vs centered gap of  $\sim 0.5$  —the gap is mainly driven by mean shift.

**Ait-Saada & Nadif 2023 [9] (ACL Short):** Counter-consensus balance —measured anisotropy and clustering NMI are **near-zero or weakly negatively correlated**, even claiming “fostering high anisotropy yields high-quality clustering representations” . → Note: they use NMI (external metric), NPM uses silhouette (internal metric); the two metric families differ in sensitivity to rogue-dim scale, so this counter-consensus does not directly transplant to silhouette.

**Gao et al. 2019 [2] (ICLR) + Wang et al. 2020 [3] (ICLR):** Theoretical foundations of anisotropy —softmax + weight tying + Zipf long-tail → narrow cone for low-frequency tokens. → Explains why BERT-style models (without uniformity loss) are necessarily strongly anisotropic.

## 6.5 §6 Methodological Contributions (independent of main study)

1. **Silhouette on contextual embedding is not real cluster measure:** 7 models  $\times$  8 domains  $\times$  whitening = 280 silhouette mostly  $\approx 0$ . Provides empirical evidence for limits of “silhouette + contextual embedding” combination.
2. **Diagnostic for “real subcategory signal vs anisotropy artifact” :** compare raw vs whitened silhouette difference. Whitened  $\approx 0$  → anisotropy artifact; whitened retains signal → real cluster signal. Simple and reproducible.
3. **Anisotropy bias in cross-model consistency:** raw Kendall  $W = 0.266$ , whitened  $W = 0.228$  (slightly drops; full 7-model  $\times$  5-treatment Kendall  $W$  table in reference doc § 2.8). Indicates raw “cross-model consistency” partially comes from anisotropy common pattern, not true consensus.
4. **Methodological scope confirmation:** sentence-transformers class models (uniformity-loss trained) are a reasonable basis for silhouette/Vendi measurement; BERT-style models exhibit a **systematic bias** on silhouette (raw values are dominated by anisotropy and collapse to zero after whitening), so cross-paradigm direct comparison is misleading. This finding is precisely why the v2.4 main study actively scopes itself back to sentence-transformers.

## 6.6 §6 Limitations (this standalone chapter’ s own honest boundaries)

| #     | Boundary                 | Description                                            |
|-------|--------------------------|--------------------------------------------------------|
| §6 L1 | Only 4 BERT-style models | Adding SciBERT / CaseLaw-BERT 5th-6th would strengthen |

| #     | Boundary                                                                       | Description                                                                                                            |
|-------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| §6 L2 | 5 treatments are literature-standard but not exhaustive                        | Can add more (e.g. Mahalanobis whitening, supervised whitening)                                                        |
| §6 L3 | “Anisotropy-dominated” inferred from correlation, not direct causal experiment | Causal verification needs training-time anisotropy strength control (e.g. adding contrastive head fine-tuning to BERT) |

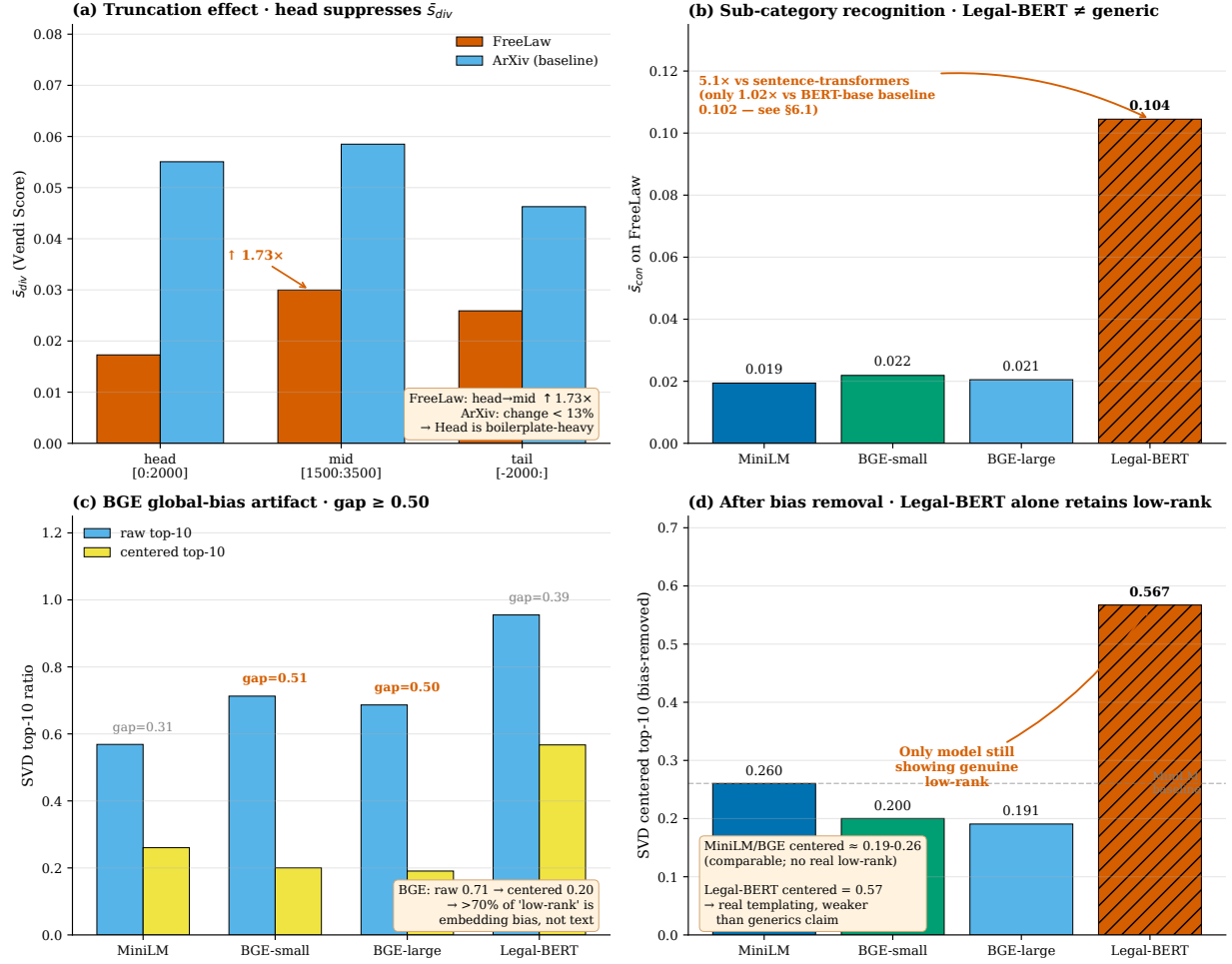


Figure 4: §6 probe summary • four-mechanism decomposition of FreeLaw “low-diversity” signal (includes Legal-BERT)

## Appendix A • Full 8-Domain Data (sentence-transformers main)

See companion data document 阶梯 1\_ 完整原始数据 \_ 含 BERT-baseline\_20260425.md:

- **§ 1 Main data** (sentence-transformers, three models)
  - § 1.2 table: 8 domains  $\times$  3 models  $\times$  4 components
- **§ 2 BERT probe data** (BERT-style, four models • for § 6 reference only)
  - § 2.2 table: 8 domains  $\times$  4 models  $\times$  4 components

## Appendix B • Main Figures

Figures are stored in the `figures/` subdirectory. Strict main-study / §6-probe split: main-study figures contain only the 3 sentence-transformers models; §6-probe figures additionally include Legal-BERT.

**PDF figure number  $\leftrightarrow$  source filename mapping** (the in-PDF “Figure 1-4” labels are auto-numbered by pandoc in order of appearance; source files keep the F1-F4 names from earlier versions for git-history continuity):

| In-PDF label | Source filename          | Section | Scope                       | Content                                                                                              |
|--------------|--------------------------|---------|-----------------------------|------------------------------------------------------------------------------------------------------|
| Figure 1     | F3_sbar_heatmap_st_§3.1  | §3.1    | sentence-transformers only  | § four components $\times$ 3 models $\times$ 8 domains heatmap                                       |
| Figure 2     | F4_spearman_st_3x3 §3.2  | §3.2    | sentence-transformers only  | one 3 $\times$ 3 pairwise Spearman matrix per component                                              |
| Figure 3     | F2_vendi_svd_st_24p §3.6 | §3.6    | sentence-transformers only  | Vendi-SVD 24-point scatter + power-law fit, $\rho \approx -0.997$                                    |
| Figure 4     | F1_I6_four_mechanisms §6 | §6      | includes Legal-BERT (probe) | 4-panel FreeLaw “low-diversity” decomposition (truncation / sub-category / BGE bias / true low-rank) |

Each figure is provided in PNG + PDF, both bilingual (EN + CN). Naming convention: F{number}\_{description}\_{cn or English}.{png,pdf}.

## Appendix C • Related Work

This appendix lists the 25 references involved in this study, organized into 6 thematic categories.

### C.1 Embedding geometry and anisotropy

(9 entries • covers anisotropy phenomena, causes, and mitigation)

- [1] **Mu, J., & Viswanath, P.** (2018). All-but-the-Top: Simple and Effective Postprocessing for Word Representations. *International Conference on Learning Representations (ICLR)*. arXiv:1702.01417.
- [2] **Gao, J., He, D., Tan, X., Qin, T., Wang, L., & Liu, T.-Y.** (2019). Representation Degeneration Problem in Training Natural Language Generation Models. *International Conference on Learning Representations (ICLR)*. arXiv:1907.12009.
- [3] **Wang, L., Huang, J., Huang, K., Hu, Z., Wang, G., & Gu, Q.** (2020). Improving Neural Language Generation with Spectrum Control. *International Conference on Learning Representations (ICLR)*.
- [4] **Ethayarajh, K.** (2019). How Contextual are Contextualized Word Representations? Comparing the Geometry of BERT, ELMo, and GPT-2 Embeddings. *Empirical Methods in Natural Language Processing (EMNLP)*. arXiv:1909.00512.
- [5] **Bis, D., Podkorytov, M., & Liu, X.** (2021). Too Much in Common: Shifting of Embeddings in Transformer Language Models and its Implications. *NAACL 2021*. ACL Anthology 2021.naacl-main.403.
- [6] **Cai, X., Huang, J., Bian, Y., & Church, K.** (2021). Isotropy in the Contextual Embedding Space: Clusters and Manifolds. *International Conference on Learning Representations (ICLR)*.
- [7] **Timkey, W., & van Schijndel, M.** (2021). All Bark and No Bite: Rogue Dimensions in Transformer Language Models Obscure Representational Quality. *Empirical Methods in Natural Language Processing (EMNLP)*. arXiv:2109.04404.
- [8] **Rudman, W., & Eickhoff, C.** (2024). Stable Anisotropic Regularization (I-STAR). *International Conference on Learning Representations (ICLR)*. arXiv:2305.19358.
- [9] **Ait-Saada, M., & Nadif, M.** (2023). Is Anisotropy Truly Harmful? A Case Study on Text Clustering. *Annual Meeting of the Association for Computational Linguistics (ACL Short)*. ACL Anthology 2023.acl-short.103.

### C.2 Diversity and quality measures

(3 entries • Vendi Score / effective rank / domain clustering)

- [10] **Friedman, D., & Dieng, A. B.** (2022). The Vendi Score: A Diversity Evaluation Metric for Machine Learning. *Transactions on Machine Learning Research (TMLR)*. arXiv:2210.02410.
- [11] **Roy, O., & Vetterli, M.** (2007). The effective rank: A measure of effective dimensionality. *15th European Signal Processing Conference (EUSIPCO)*, pp. 606-610.
- [12] **Aharoni, R., & Goldberg, Y.** (2020). Unsupervised Domain Clusters in Pretrained Language Models. *Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 7747-7763.

### C.3 Scaling Laws and data quality

(3 entries • reference for subsequent interventional studies)

- [13] **Subramanyam, S., Chen, Y., & Grossman, S.** (2025). Quality-aware scaling laws for language model pretraining. *arXiv:2510.03313v2*.
- [14] **Bahri, Y., Dyer, E., Kaplan, J., Lee, J., & Sharma, U.** (2024). Explaining Neural Scaling Laws. *Proceedings of the National Academy of Sciences (PNAS)*, 121(27): e2311878121.
- [15] **Li, J., Fang, A., Smyrnis, G., et al.** (2024). DataComp-LM: In search of the next generation of training sets for language models. *NeurIPS Datasets and Benchmarks (DCLM)*. *arXiv:2406.11794*.

#### C.4 Sentence Embedding models

(4 entries · original papers of the three models used in main §1-§5)

- [16] **Reimers, N., & Gurevych, I.** (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *Empirical Methods in Natural Language Processing (EMNLP)*. *arXiv:1908.10084*.
- [17] **Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., & Zhou, M.** (2020). MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. *NeurIPS 2020*. *arXiv:2002.10957*.
- [18] **Xiao, S., Liu, Z., Zhang, P., & Muennighoff, N.** (2023). C-Pack: Packaged Resources To Advance General Chinese Embedding (BGE / FlagEmbedding). *arXiv:2309.07597*.
- [19] **Liu, Z., Shao, S., Shi, X., Lian, D., & Xiao, S.** (2022). RetroMAE: Pre-Training Retrieval-oriented Language Models Via Masked Auto-Encoder. *EMNLP 2022*. *arXiv:2205.12035*.

#### C.5 BERT-style domain-specific models (§6 references)

(4 entries · original papers of probe models used in §6 standalone chapter)

- [20] **Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K.** (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL 2019*, pp. 4171-4186. *arXiv:1810.04805*.
- [21] **Chalkidis, I., Fergadiotis, M., Malakasiotis, P., Aletras, N., & Androutsopoulos, I.** (2020). LEGAL-BERT: The Muppets straight out of Law School. *Findings of EMNLP 2020*. *arXiv:2010.02559*.
- [22] **Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J.** (2020). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4): 1234-1240. *arXiv:1901.08746*.
- [23] **Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., Naumann, T., Gao, J., & Poon, H.** (2021). Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing (PubMedBERT). *ACM Transactions on Computing for Healthcare*, 3(1): 1-23. *arXiv:2007.15779*.

#### C.6 NPM framework itself

(2 entries · prior NPM-series releases on Zenodo)

[24] **Ding, T.** (2026). Neural Percolation Model: A Porous-Media-Inspired Framework for Neural Network Training Dynamics. Zenodo. DOI: 10.5281/zenodo.19209722.

[25] **Ding, T.** (2026). Cross-Family Convergence of Neural Network Weight Skeletons (NPM-K). Zenodo. DOI: 10.5281/zenodo.19652706.

## **Appendix D • Code and Data Availability**

All code, data, and a transparent data-reference document for this study are publicly available on GitHub:

**Code & data:** <https://github.com/tiexinding/data-quality-vec-public> (release: v2.4 / 2026-04-25)

**Zenodo DOI:** [10.5281/zenodo.19762059](https://doi.org/10.5281/zenodo.19762059) —please use this DOI when citing this work.

Repository contents: bilingual technical report v2.4 (EN + CN) + 19 CSV/JSON data files (primary / probe / ablation / 5-treatment) + F1-F4 main figures (16 PNG/PDF, EN + CN) + main pipeline script (`run_stage1_sbar_v2.py`) + SVD anisotropy diagnostic + K-scan + figure generation + PDF builder. License: MIT.

Intermediate embeddings (`.npy`, ~380 MB) and text caches (~137 MB) are NOT included in the repository; they will be regenerated on the fly from `monology/pile-uncopyrighted` (HuggingFace) and the listed model checkpoints by `code/run_stage1_sbar_v2.py`.

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**Version:** v2.4 • 2026-04-25 **Release intent:** Zenodo deposit • main scope strictly sentence-transformers • §6 standalone methodological contribution